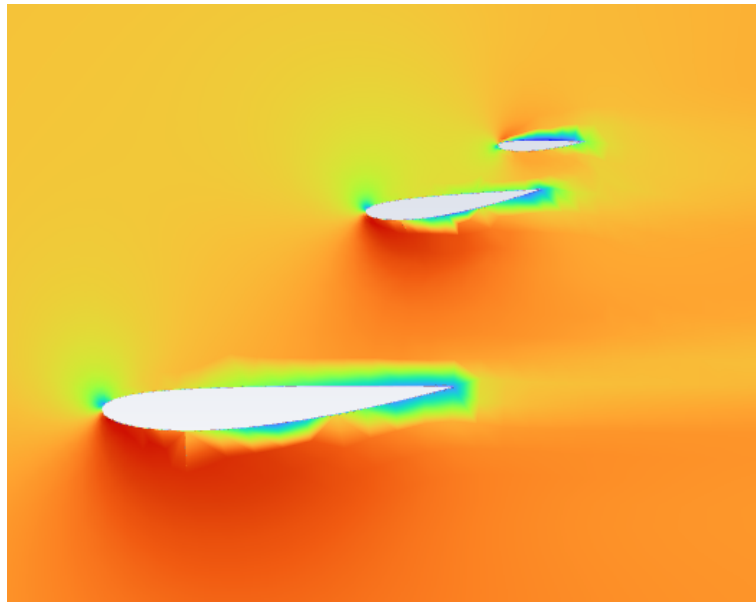


Aerodynamic Optimisation of a Two-Element Wing Configuration

Assignment 3 – Maximum Downforce Study

Mukundan Subbaiya | Neha Mandapaka



Introduction

Multi-element wing systems are fundamental to modern motorsport aerodynamics. By splitting a single lifting surface into two elements separated by a carefully designed slot, the effective maximum lift coefficient can substantially exceed that of any single element operating in isolation. The slot acts as a conduit for high-momentum fluid, re-energising the boundary layer on the suction surface of each downstream element and delaying separation to higher angles of attack [1]. In a race car context, the wing is inverted so that the aerodynamic force acts downward, increasing tyre contact forces and cornering capability.

Optimising such systems is a non-trivial task: the relevant design variables – element angles, gap distances, and relative positions – interact nonlinearly, and the flow physics involve adverse pressure gradients, boundary layer development, and potential separation. Traditional parametric sweeps are computationally prohibitive for multi-variable problems. Design of Experiments combined with Response Surface Methodology provides an efficient alternative: a small number of carefully chosen simulations is used to construct a mathematical surrogate (meta-model) of the aerodynamic response, which can then be searched by an optimisation algorithm at negligible computational cost.

The aim of this work is to apply a full DoE–RSM–MOGA optimisation workflow in ANSYS Workbench and DesignXplorer to a two-element inverted wing operating at 70 km/h. The specific objectives are: (i) to select and justify three geometrical design parameters on aerodynamic grounds; (ii) to construct and validate a response surface for C_L and C_D ; (iii) to identify the optimal configuration using MOGA; and (iv) to verify the prediction by direct CFD simulation and explain the flow physics driving the performance improvement.

Method

The methodology follows the **ANSYS DesignXplorer** workflow: parametric geometry → mesh → CFD → DoE → response surface → optimisation → verification.

Geometry and Baseline Configuration

The two-dimensional geometry consists of two inverted airfoils in tandem, using the same profile as Assignment 1. The main element (element 1) has chord $c_1 = 0.4$ m and the secondary element (element 2) has $c_2 = 0.2$ m. Both airfoils are mirrored about the chord line so that the cambered surface faces downward, generating downforce rather than lift. The freestream conditions are $V_\infty = 19.44$ m/s (70 km/h), $\rho = 1.205$ kg/m³, and $\mu = 1.82 \times 10^{-5}$ kg/(m·s), giving a Reynolds number $Re = \rho V_\infty c_1 / \mu = 5.15 \times 10^5$.

The baseline configuration follows the assignment specification: $AoA_1 = 1^\circ$, $AoA_2 = 4^\circ$, $TranslateY = 0.15$ m. The geometry was constructed in ANSYS DesignModeler by importing the airfoil coordinates via the *Curve from Points* feature and applying successive body transformations: mirror (for downforce orientation), scale (to target chord), rotate (for AoA), and translate (for positioning). The fluid domain is 25×25 m with a 7.5×7.5 m Body of Influence (BOI) surrounding the airfoils. Named selections were defined for the inlet (velocity inlet), outlet (pressure outlet), far-field top/bottom (symmetry), and both airfoil surfaces (no-slip walls). Each body transformation angle and translation was exposed as a Workbench input parameter using the dimension checkbox in DesignModeler, enabling automatic geometry regeneration for each DoE design point.

Design Parameter Selection

Three parameters were selected for the optimisation study. Selection was based on aerodynamic sensitivity and physical relevance to the downforce objective.

1. **AoA₁** – **Main element angle of attack (0° to 8°)**: The main element carries most of the aerodynamic load due to its larger chord. Increasing AoA₁ steepens the pressure gradient on the suction surface, augmenting the bound circulation and downforce per the Kutta–Joukowski theorem. Precursor simulations confirmed that the k – ω SST model remains reliable up to 8°; beyond this, large-scale separation develops and steady-state RANS diverges.
2. **AoA₂** – **Secondary element angle of attack (0° to 7°)**: The secondary element modulates the rear circulation and acts analogously to a flap, augmenting element 1 circulation through the pressure interaction described by Smith [1]. The upper bound was reduced to 7° following precursor simulations in which Fluent failed to converge at AoA₂ $\geq$ 7.5° due to onset of full separation on element 2.
3. **TranslateY** – **Vertical gap (0.1 m to 0.40 m)**: This parameter controls the slot width between the elements and hence the mass flow rate of the re-energising slot jet. The lower bound of 0.10 m was set to guarantee geometric clearance at all AoA combinations in the DoE, confirmed by testing the most extreme factorial corner point before running the full DoE.

Mesh Generation and Independence Study

An unstructured triangular mesh with an inflation layer on all airfoil surfaces was generated in the ANSYS Mesher. The first-cell height was set to $y_1 = 1.5 \times 10^{-5}$ m, computed from the flat-plate turbulent boundary layer estimate for $y^+ = 1$:

$$y_1 = \frac{y^+ \mu}{\rho u_\tau}, \quad u_\tau = \sqrt{\frac{\tau_w}{\rho}}, \quad \tau_w = \frac{1}{2} \rho V_\infty^2 C_f \quad (1)$$

where $C_f = 0.058 Re^{-0.2} = 0.00418$ for the turbulent flat plate. This yields $u_\tau = 0.889$ m/s and $y_1 = 1.7 \times 10^{-5}$ m; 1.5×10^{-5} m was used to provide a conservative margin. The inflation layer comprised 15 layers with a growth rate of 1.2. Mesh independence was verified on the baseline configuration using three refinement levels:

Table 1: Mesh independence study. ΔC_L is the relative change with respect to the fine mesh.

Mesh	Cells	C_L	C_D	ΔC_L
Coarse	$\approx 75,000$	−0.7041	0.03089	4.7%
Medium	$\approx 210,000$	−0.7343	0.02852	0.6%
Fine	$\approx 380,000$	−0.7386	0.02803	—

The relative change in C_L between the medium and fine meshes was 0.6%, below the 2% independence criterion. The medium mesh was therefore used for all DoE simulations, reducing computational cost by approximately 80% relative to the fine mesh.

CFD Solver Settings

All simulations used the following settings:

- **Solver:** Pressure-based, steady-state .
- **Turbulence model:** $k-\omega$ SST [2]. Selected because it blends the near-wall accuracy of $k-\omega$ with the freestream robustness of $k-\epsilon$, and it is specifically formulated to handle adverse pressure gradient flows with potential trailing-edge separation on both airfoil elements.
- **Spatial discretisation:** SIMPLE pressure-velocity coupling; second-order upwind for all transport equations.
- **Boundary conditions:** Velocity inlet ($V_\infty = 19.44$ m/s, turbulence intensity 0.1%, viscosity ratio 1); pressure outlet (0 Pa gauge); symmetry on top/bottom domain edges; no-slip wall on all airfoil surfaces.
- **Reference values:** Area = 0.4 m^2 , $V_{\text{ref}} = 19.44$ m/s, $\rho = 1.205 \text{ kg/m}^3$, with C_l and C_d defined as:

$$C_L = \frac{F_Y}{\frac{1}{2}\rho V_\infty^2 c_1}, \quad C_D = \frac{F_X}{\frac{1}{2}\rho V_\infty^2 c_1} \quad (2)$$

- **Convergence:** All residuals $< 10^{-6}$, and stabilisation of C_L and C_D force monitors confirmed.
- **Output parameters:** Fluent Report Definitions for C_L , C_D , and their corresponding dimensional forces were created with the *Create Output Parameter* option, enabling automatic recording by DesignXplorer at each design point.

Design of Experiments

Optimal Space-Filling Design (OSF) was selected for the DoE. OSF distributes design points across the parameter space using a Max-Min Distance criterion, maximising the minimum distance between any two points so that the entire design space is sampled as uniformly as possible [5]. Unlike Central Composite Design, which places points at pre-defined factorial corners and axial extremes, OSF imposes no assumption on the shape of the response. This makes it more appropriate for aerodynamic optimisation problems where the lift and drag responses are nonlinear functions of angle of attack - particularly near the onset of boundary layer separation - and where the optimal response surface type is not known before the simulations are run. For three parameters, DesignXplorer generated $5 \times n_{\text{params}} = 15$ design points by default, which exceeds the minimum of $2 \times n_{\text{params}} = 6$ points required to support a reliable response surface and provides good interior coverage of the design space. The parameter bounds used are summarised in Table 2.

Table 2: Design parameter bounds used in the OSF.

Parameter	Symbol	Lower bound	Baseline	Upper bound
Main element AoA	AoA ₁	0°	1°	8°
Secondary element AoA	AoA ₂	0°	4°	7°
Vertical gap	TranslateY	0.1 m	0.15 m	0.40 m

All 15 design points converged successfully.

Response Surface Methodology

The response surface was built using *Genetic Aggregation*. This method selects and combines surface types (polynomial, Kriging, neural network, and non-parametric regression) to minimise cross-validation error for each output independently. This model was chosen over a fixed polynomial because a pure quadratic model would not adequately represent the nonlinear drag response near the separation boundary. To evaluate surface quality, R^2 was calculated for both learning points (fit quality on training data) and cross-validation points (predictive accuracy on unseen data). The relative maximum and average absolute cross-validation errors were analysed to identify impaired local predictions.

Optimisation

The Multi-Objective Genetic Algorithm (*MOGA*), a variant of NSGA-II using controlled elitism [3], was applied to simultaneously minimise C_D (drag) and minimise C_L (since $C_L < 0$ for downforce, minimising C_L maximises the downforce magnitude). No additional constraints were imposed. MOGA was configured with 3000 initial samples, 600 samples per iteration, and a maximum of 20 iterations. MOGA was selected over gradient-based methods (e.g. NLPQL) because it is robust to local optima on a nonlinear response surface and handles competing objectives by resolving the full Pareto front. The optimal candidate was subsequently verified by a direct Fluent simulation at the exact recommended parameter values.

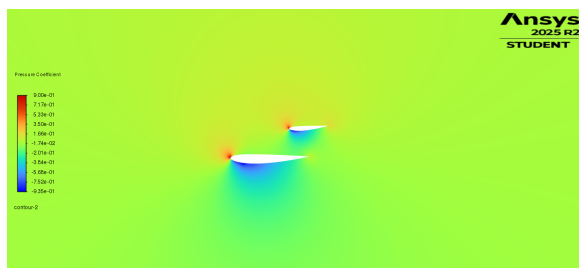
Results

Baseline Configuration

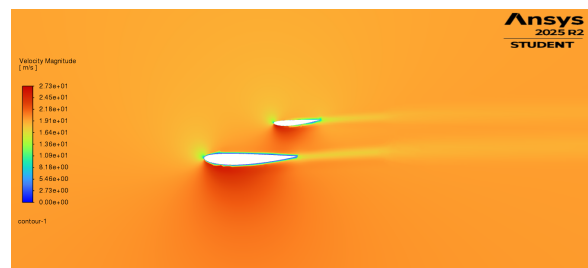
The baseline simulation ($AoA_1 = 1^\circ$, $AoA_2 = 4^\circ$, $TranslateY = 0.15$ m) converged in approximately 400 iterations. The aerodynamic coefficients and forces are:

$$C_L = -0.7343, \quad C_D = 0.02852, \quad L/D = 25.74, \quad F_L = 66.9 \text{ N/m}, \quad F_D = 2.60 \text{ N/m}$$

The negative C_L confirms the downforce direction. The pressure coefficient distribution and velocity contour for the baseline are shown in Figures 1a and 1b.



(a) Pressure coefficient distribution on both elements. The suction peak on element 1 is at $x/c \approx 0.15$.



(b) Velocity magnitude contour. The slot jet between the elements is visible as a localised high-velocity region.

Figure 1: Baseline configuration ($AoA_1 = 1^\circ$, $AoA_2 = 4^\circ$, $TranslateY = 0.15$ m).

DoE Results

Table 3 presents the full 15-point OSF results. $|C_L|$ ranges from 0.046 to 0.551 and C_D from 0.026 to 0.052, demonstrating that the chosen parameters produce meaningful aerodynamic variation across the design space.

Table 3: OSF design point results. C_L is negative for downforce; $|C_L|/C_D$ is the aerodynamic efficiency.

DP	AoA ₁ (°)	AoA ₂ (°)	TranslateY (m)	C_D	C_L	$ C_L /C_D$
1	1.33	1.40	0.133	0.02780	−0.096	3.5
2	5.07	1.00	0.120	0.04316	−0.182	4.2
3	7.20	2.20	0.253	0.04874	−0.354	7.3
4	1.87	5.40	0.240	0.02777	−0.091	3.3
5	7.73	5.00	0.213	0.05182	−0.551	10.6
6	0.27	3.80	0.173	0.02721	−0.111	4.1
7	2.93	4.20	0.107	0.03085	−0.118	3.8
8	0.80	2.60	0.267	0.02594	−0.114	4.4
9	3.47	3.40	0.187	0.03288	−0.142	4.3
10	4.00	1.80	0.293	0.03144	−0.173	5.5
11	6.13	0.20	0.200	0.04420	−0.205	4.6
12	6.67	3.00	0.147	0.04356	−0.345	7.9
13	4.53	5.80	0.160	0.03504	−0.344	9.8
14	2.40	0.60	0.227	0.02841	−0.046	1.6
15	5.60	4.60	0.280	0.03821	−0.389	10.2

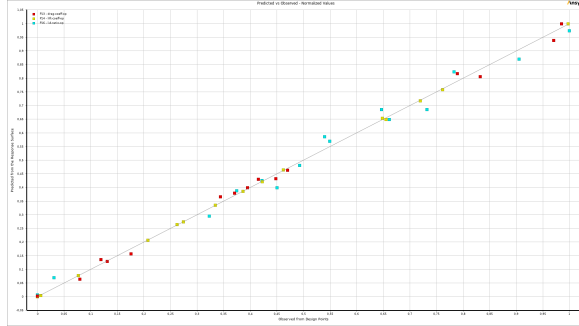
Response Surface Quality

Table 4 summarises the Goodness-of-Fit metrics for the C_D and C_L response surfaces. The cross-validation metrics are the primary quality indicators.

Table 4: Response surface Goodness-of-Fit metrics (Genetic Aggregation method).

Metric	C_D surface		C_L surface	
	Learning	Cross-val.	Learning	Cross-val.
R^2 (best = 1)	0.9968	0.9742	0.9999	0.9995
RMSE (best = 0)	0.000537	0.001526	0.00202	0.00486
Max relative error (%)	9.90	28.98	2.23	4.71
Average relative error (%)	4.68	13.08	0.67	1.79

The three-dimensional response surface for $|C_L|$ and the local sensitivity chart are shown in Figure 2.



(a) Predicted V/S Observed Surface Response



(b) Local sensitivity chart showing the relative contribution of each parameter to C_L and C_D variance. AoA_1 dominates the C_L response.

Figure 2: Response surface and local sensitivity analysis.

Optimisation Results and Verification

MOGA converged after 5584 evaluations across 20 iterations. Three candidate points were returned. Candidate 3 was selected on the basis of the lowest C_D at near-equal C_L . Table 5 presents all three candidates alongside the verified simulation result.

Table 5: MOGA candidate points and direct simulation verification for the selected optimal (Candidate 3).

Parameter / Output	Candidate 1	Candidate 2	Candidate 3	Verified
AoA_1 ($^\circ$)	3.87	3.60	3.68	3.68
AoA_2 ($^\circ$)	7.09	7.49	7.31	7.31
TranslateY (m)	0.2245	0.2249	0.2246	0.2246
C_D (predicted / actual)	0.03872	0.03869	0.03858	0.03623
C_L (predicted / actual)	-1.125	-1.124	-1.122	-1.1319
Δ_{C_L}	—	—	—	0.88%
Δ_{C_D}	—	—	—	6.10%

The complete baseline-to-optimal comparison is given in Table 6.

Table 6: Aerodynamic performance: baseline vs. verified optimal configuration. Forces are per unit span, $q = \frac{1}{2}\rho V_\infty^2 = 227.7$ Pa.

Metric	Baseline	Optimal	Δ (abs.)	Change
$ C_L $ (downforce)	0.7343	1.1319	+0.3976	+ 54.2%
C_D (drag)	0.02852	0.03623	+0.0077	+27.0%
L/D ratio	25.74	31.24	+5.50	+ 21.4%
F_L (N/m)	66.9	103.1	+36.2	+ 54.2%
F_D (N/m)	2.60	3.30	+0.70	+27.0%
AoA_1	1.0 $^\circ$	3.68 $^\circ$	+2.68	—
AoA_2	4.0 $^\circ$	7.31 $^\circ$	+3.31	—
TranslateY	0.15 m	0.225 m	+0.075 m	—

The flow field at the verified optimal configuration is shown in Figures 3 and 4.

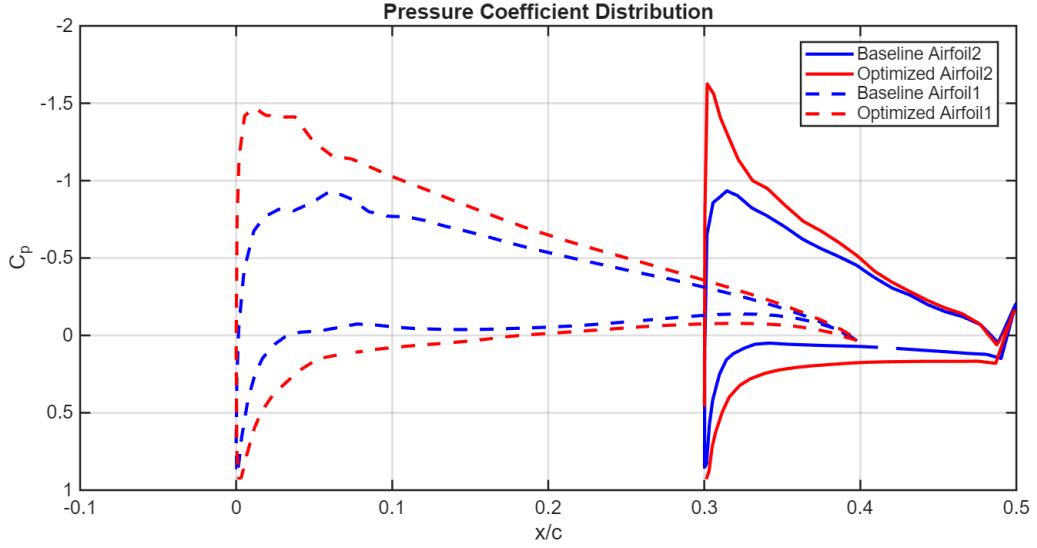


Figure 3: Pressure coefficient distributions for the baseline (Blue) and optimal (Red) configurations on element 1 (dashed) and element 2 (solid).

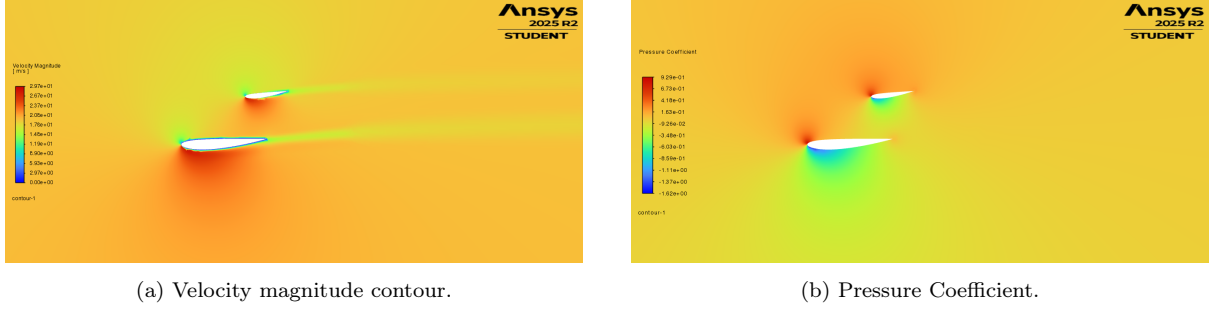


Figure 4: Verified optimal configuration flow field ($AoA_1 = 3.68^\circ$, $AoA_2 = 7.31^\circ$, $TranslateY = 0.225$ m).

Discussion

Physical Interpretation of the Optimal Configuration

The optimiser converged to $AoA_1 = 3.68^\circ$ and $AoA_2 = 7.31^\circ$, both substantially larger than the baseline values of 1° and 4° , with a 50% wider vertical gap. The 54.2% downforce improvement and 21.4% L/D improvement are driven by two coupled aerodynamic mechanisms.

Circulation augmentation. A higher AoA_1 steepens the adverse pressure gradient on the suction surface of the main element, increasing its bound circulation. Element 2 simultaneously induces an upwash on the trailing region of element 1, further augmenting its effective circulation beyond what either element could achieve alone [1]. This is visible in Figure 3 as a significantly deeper suction peak on element 1. The sensitivity analysis (Figure 2) confirms that AoA_1 is the dominant driver of C_L variance, consistent with the Kutta–Joukowski scaling with chord: the same angular increment on the larger-chord element 1 generates twice the circulation change of element 2.

Slot-jet boundary layer re-energisation. The wider vertical gap at the optimal configuration (0.225 m vs. 0.15 m baseline) increases the mass flow through the slot. The accelerated jet, visible in Figure 4, re-energises the boundary layer on the suction surface of element 2,

delaying separation. At higher operating angles the adverse pressure gradient on element 2 is steeper and demands a stronger jet to avoid stall. The wider gap provides this at a modest drag cost.

Drag trade-off. The 27.0% C_D increase is attributable primarily to increased pressure drag at higher AoA. The stagnation point migrates further along the pressure surface, increasing the projected frontal area and the pressure difference across both elements. Nevertheless, the net L/D improvement of 21.4% confirms that the downforce gain outweighs the drag penalty in efficiency terms. From a race car perspective this trade-off is favourable: downforce directly improves cornering acceleration, whereas drag primarily reduces straight-line top speed.

Response Surface Quality Assessment

The Cross-validation $R^2 = 0.9995$, average error of 1.79%, and maximum error of 4.71%, indicate the exceptional reliability of the CL surface. A CL prediction error of 0.88% was confirmed in the verification simulation. The CD surface ($R^2 = 0.974$, average error 13.1%) is well-fitted except for a 28.98% localised error due to the nonlinear drag response in the high-AoA region where separation begins. At the optimal point, the CD was 6.1% lower than predicted (0.03623 vs. 0.03858), indicating that the response surface conservatively over-predicted drag. This result was more favourable than the optimiser expected. Since the L/D ratio surface exhibited poor cross-validation quality ($R^2 = 0.72$) due to nonlinear amplification of Cd errors, it was not used as an optimisation objective. Instead, it was used as a post-processing metric from verified simulation values.

MOGA Method Evaluation

MOGA was well-suited to this problem for three reasons. First, the competing objectives form a natural Pareto front that MOGA resolves explicitly, without requiring a weighted combination of objectives that would bias the search. Second, the nonlinear response surface precluded gradient-based methods such as NLPQL, which are susceptible to local optima near inflection points. Third, convergence after 5584 evaluations in 20 iterations, with the three candidates showing less than 0.4% variation in both C_L and C_D , indicates a well-defined narrow optimum.

A limitation of the MOGA approach is that it searches the response surface rather than the true objective function. Localised inaccuracies in the C_D surface (maximum error 28.98%) could in principle bias the optimal candidate toward regions where the surface under-predicts drag. This is exactly why candidate verification is mandatory and non-negotiable. The 6.1% C_D verification error is acceptable but underscores the importance of direct simulation at the recommended optimal point.

Limitations of the Present Approach

Several modelling assumptions limit generalisability:

- **Two-dimensional analysis.** The simulations are strictly 2D and cannot capture spanwise flow, tip vortices, or end-plate effects. The C_L values reported represent an infinite-span upper bound and would be reduced for a finite wing by the induced drag and tip leakage effects.

- **Steady-state RANS.** The k - ω SST model solves time-averaged equations, which becomes increasingly approximate at high AoA where low-frequency unsteady vortex shedding from the trailing edges develops. At the optimal $\text{AoA}_2 = 7.31^\circ$, the flow is marginally attached and unsteady effects could cause the true time-averaged C_L to differ from the steady prediction by up to 5–10%.
- **Response surface extrapolation.** The MOGA candidates had AoA values slightly beyond the original OSF bounds, relying on polynomial extrapolation. The close verification agreement (0.88% error) confirms the extrapolation was benign in this case, but additional design points in the extrapolated region would further strengthen confidence.
- **Fixed chord lengths.** Chord lengths were fixed at $c_1 = 0.4$ m and $c_2 = 0.2$ m. Allowing chord variation in a future study could yield further gains, though the risk of geometric intersection across all DoE combinations would require careful range management.

Conclusions

A DoE–RSM–MOGA optimisation workflow was applied to a two-element inverted wing at 70 km/h. The following conclusions are drawn:

1. **AoA₁ is the dominant parameter** for downforce, confirmed by the local sensitivity analysis, consistent with Kutta–Joukowski lift scaling with chord length.
2. **The response surface quality was excellent for C_L** (cross-validation $R^2 = 0.9995$, max error 4.71%) and good for C_D ($R^2 = 0.974$, max error 28.98%).
3. **The verified optimal configuration** ($\text{AoA}_1 = 3.68^\circ$, $\text{AoA}_2 = 7.31^\circ$, $\text{TranslateY} = 0.225$ m) achieves **54.2% more downforce** and a **21.4% higher L/D ratio** compared to the baseline.
4. **Response surface prediction accuracy** at the optimal point was 0.88% for C_L and 6.1% for C_D , validating the Genetic Aggregation meta-model.
5. **Steady 2D RANS provides a reliable and efficient basis** for preliminary multi-element wing optimisation, with the caveats that 3D spanwise effects and near-stall unsteadiness require higher-fidelity modelling for final design.

References

- [1] Smith, A.M.O. (1975). *High-lift aerodynamics*. Journal of Aircraft, 12(6), 501–530.
- [2] Menter, F.R. (1994). *Two-equation eddy-viscosity turbulence models for engineering applications*. AIAA Journal, 32(8), 1598–1605.
- [3] Deb, K., Pratap, A., Agarwal, S., and Meyarivan, T. (2002). *A fast and elitist multiobjective genetic algorithm: NSGA-II*. IEEE Transactions on Evolutionary Computation, 6(2), 182–197.
- [4] van Dam, C.P. (2002). *The aerodynamic design of multi-element high-lift systems for transport airplanes*. Progress in Aerospace Sciences, 38(2), 101–144.
- [5] ANSYS Inc. (2023). *ANSYS Fluent Theory Guide, Release 2023 R1*. ANSYS Inc., Canonsburg, PA.
- [6] Andersson, N. (2021). *Turbulence Modeling for CFD*. Lecture notes, Division of Applied Thermodynamics and Fluid Mechanics, Linköpings Universitet.